CLOUD & DEVOPS

Terraform & IaC Visual Guide

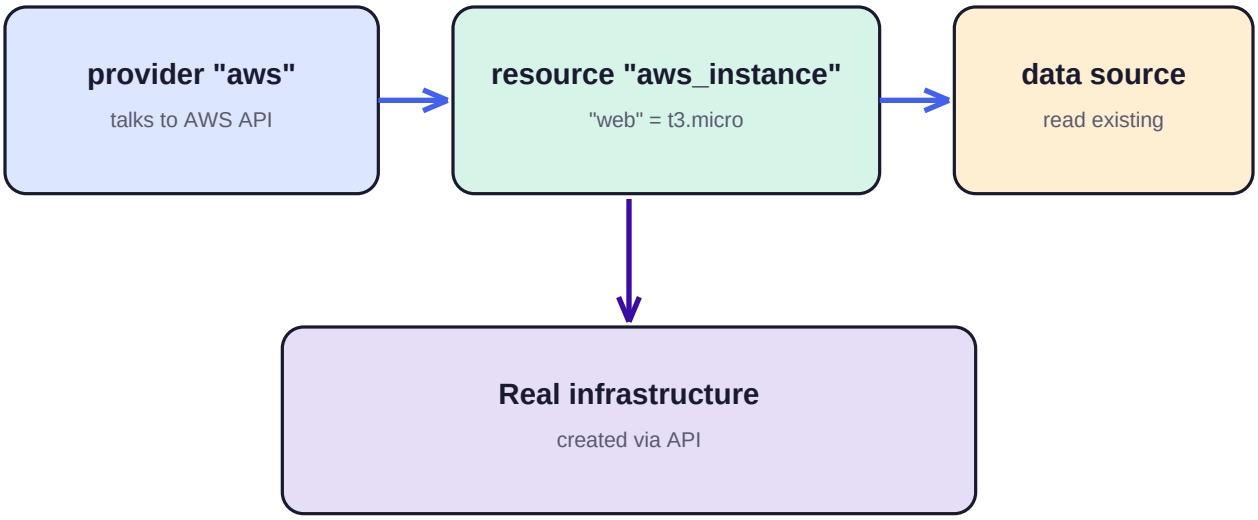
Plan, apply, repeat —
infrastructure as reviewable code.

ANIMATED WALKTHROUGH
Step through pages to play the animation

STEP 1/6

Providers & Resources

Providers talk to APIs; resources declare what should exist.

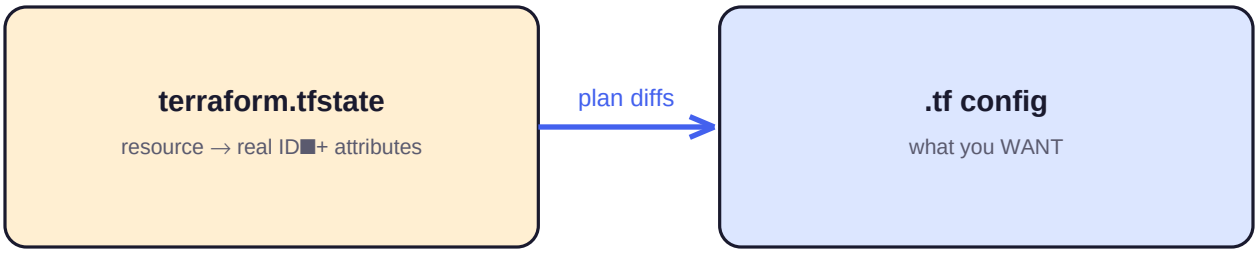


Declarative: you describe the END state. Terraform figures out the create/update/delete order.

STEP 2/6

The State File

Terraform's memory: what it created, mapped to real IDs.



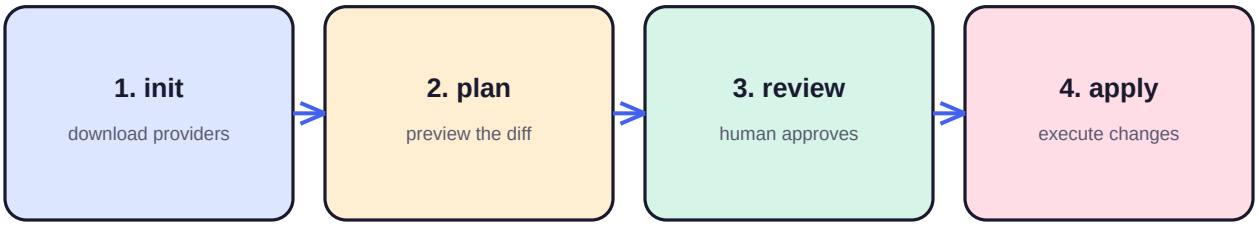
Plan compares config vs state vs reality → proposes a change set.

- State is the source of truth for Terraform — lose it and it will try to recreate everything.
- Never hand-edit state; use `terraform import` and `terraform state mv` instead.
- State can hold secrets — encrypt it and restrict access.

STEP 3/6

The Plan / Apply Cycle

Review the diff, then let Terraform execute it.



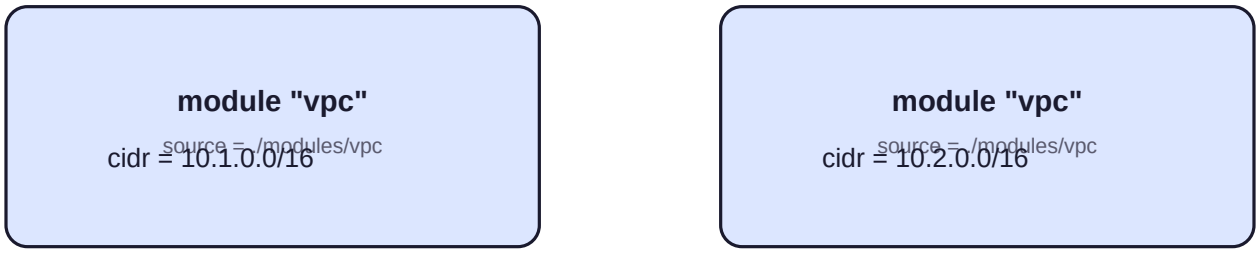
CI runs plan on every PR and posts the diff as a comment — infrastructure gets code review.

`terraform plan -out=tfplan` + `terraform apply tfplan`: apply EXACTLY what was reviewed, nothing more.

STEP 4/6

Modules: Reusable Building Blocks

Package common patterns once; instantiate with different inputs.



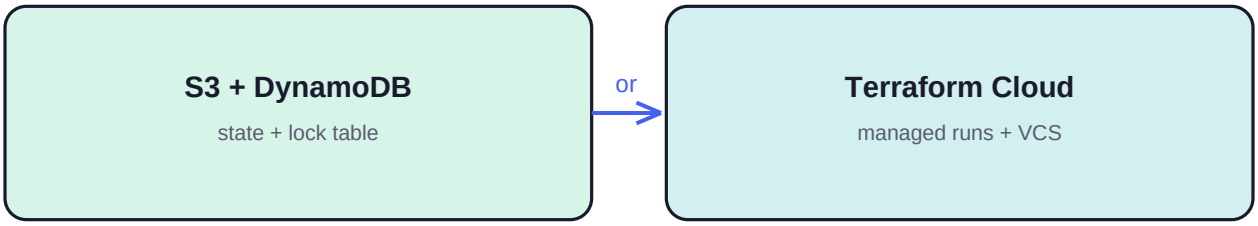
Same module, different inputs: dev and prod networks from one tested codebase.

- Inputs (variables) in, outputs out — modules are functions for infrastructure.
- Pin module versions (git tags / registry versions) — floating refs break prod at 2am.
- Compose modules: network → cluster → apps, each with a clear contract.

STEP 5/6

Remote Backends & Locking

Shared state needs a backend with locking — not a laptop file.



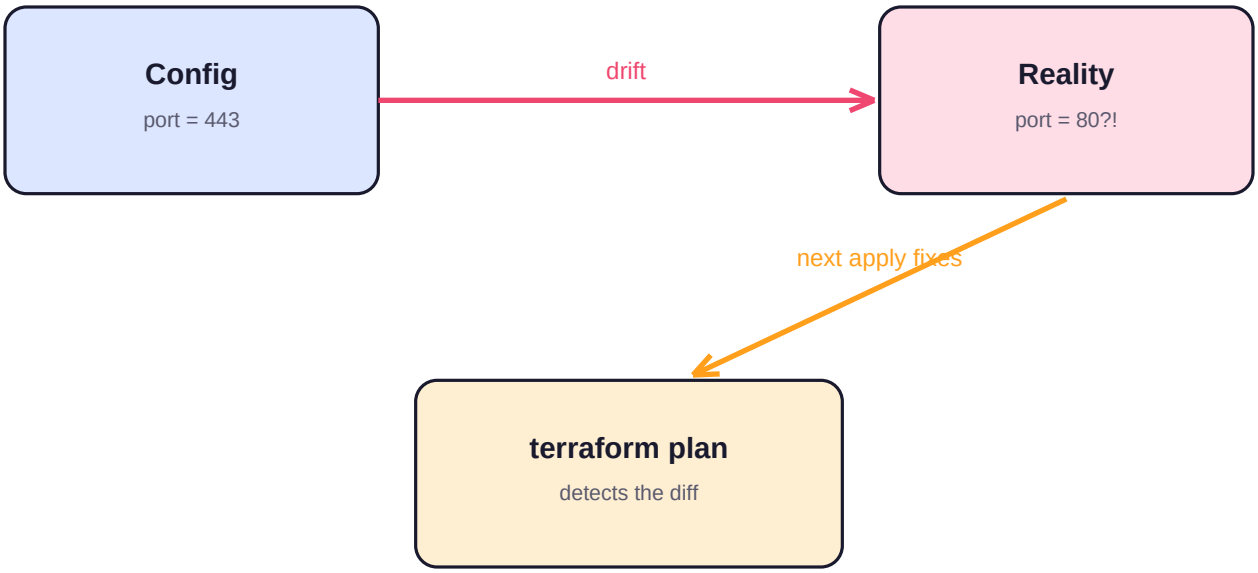
Two engineers apply at once without locking → corrupted state, duplicated resources.

- Backend stores state remotely; the lock prevents concurrent writes.
- Enable versioning on the state bucket — a bad write is recoverable.
- Separate state per environment (dev/stage/prod): smaller blast radius.

STEP 6/6

Drift: When Reality Diverges

Hand-changes in the console create drift — plan finds it.



Schedule nightly `plan` in CI: alert on unexpected diffs — that's your drift detector.

STEP ✓/✓

Key Takeaways

Terraform & IaC — Remember

- Declarative config + provider APIs: describe the end state, Terraform orders the steps.
- The state file is Terraform's memory — back it up, lock it, never hand-edit it.
- Plan → review → apply: infrastructure gets the same PR review as code.
- Modules are functions for infra: inputs in, outputs out, versions pinned.
- Remote backends + locking make teamwork safe; nightly plans catch drift.

Keep exploring

Pair this guide with the matching video walkthroughs
and hands-on labs at interstitiumlabs.dev